

Course: B. Tech.

Semester : IV

Branch : Electronics & Computer Engineering/Electronics.& Computer Science

Subject Code & Name: BTECPE405D - Linux OS

Max Marks: 60

Date: 22/07/2023

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Answer any two of the following.		12
A) Explain Architecture of Linux in detail .	CO 01	6
B) Define Linux. Also explain different features of Linux.	CO 01	6
C) Differentiate between the following terms. i) printf() & fprintf() ii) scanf() & fscanf()	CO 01,02	6
Q.2 Answer any two of the following.		12
A) Write a shell script that takes a name of a folder, and delete all sub folders of size 0?	CO 03	6
B) What do you mean by a Process States in Linux?	CO 03,04	6
C) What is Linux Shell? What type of Shells are there in Linux?	CO 02,03	6
Q.3 Answer any two of the following.		12
A) Differentiate between shell variables and environment variables and user defined variables.	CO 02	6
B) Explain the following concepts about pipes: i) Pipes between two process. ii) Pipes among three processes in a shell.	CO 04	6
C) With suitable example explain grep commands used for i) Case insensitive search	CO 02,03	6

- ii) Displaying the count of number of matches
- iii) Display the file names that matches the pattern

Q. 4 Answer any two of the following. 12

- A) Define a system call? Explain how the system call differs from that of the library functions CO 05 6
- B) Describe the following commands with example for each. CO 02, 03 6
 - i) mkdir ii) rmdir iii) chdir iv) getcwd
- C) Write a shell program for counting characters, words and line? CO 02 6

Q. 5 Answer any two of the following. 12

- A) Explain 'creating lock files', 'locking regions' and 'use of read and write with locking'. CO 03 6
- B) Briefly describe deadlocks. CO 03 6
- C) Write a shell script to find out the factorial of a number. CO 04, 05 6

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